Portfolio

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Art Beyond Observation

Museums are often silent, solemn places where we stand at a distance and observe, yet this traditional approach often limits our emotional engagement and comprehension. But what if a painting could speak back? **What if your very presence could bring life into a static canvas**, revealing the voices, conflicts, and emotions of those frozen in time? How does our connection to art change when we stop being eyes without bodies and start participating in the world behind the frame?

Art Beyond Observation is an interactive museum installation that **transforms passive viewing into an active, embodied exploration of an artwork's context**. Using a system of ”invisible” ultrasonic sensors and spatialized audio, the project un-silences historical paintings by translating a visitor's physical movement and proximity into a location-aware narrative. By turning the visitor’s body into the primary interface, the project connects the painting, the physical object itself, to its social reality, developing historical empathy and a deeper emotional connection to art.

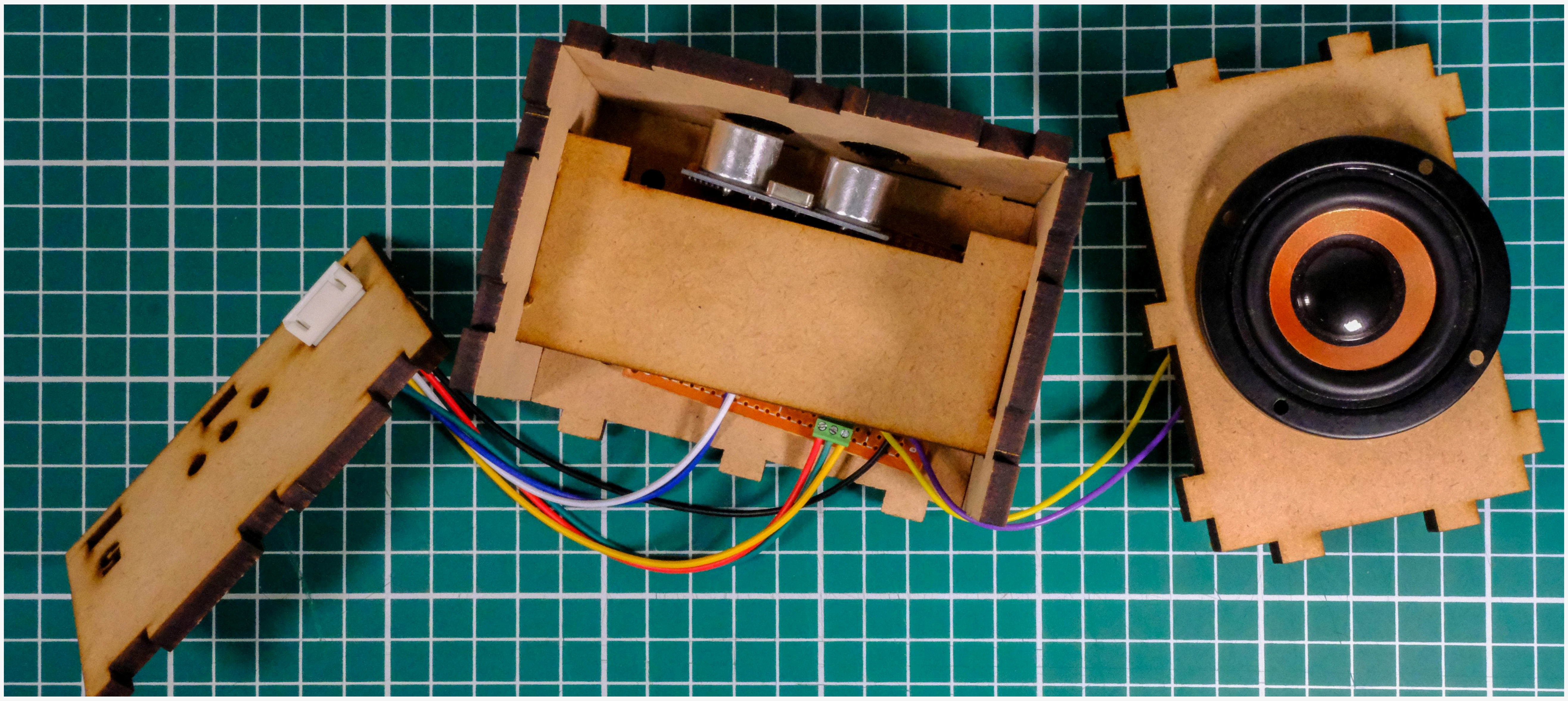
Users listen to the voices of the elements on the painting that reflect their physical presence in front of it.

Project information:

2025 | Individual work | Interactive installation built with Arduino

Master’s Final Project prepared for the attainment of the degree of Master in Interaction Design

The painting is divided in 5 groups, each one with 3 or 2 zones. When the visitor steps in one of these zones, the system plays the audio files of the character that corresponds to their physical position, but on the frame.



Detail of one of the project's boxes, showcasing the hardware inside. It's made of a ultrasonic sensor, a soldered board with a Micro SD card reader and a sound box. This allowed the system to be cheap and customizable.

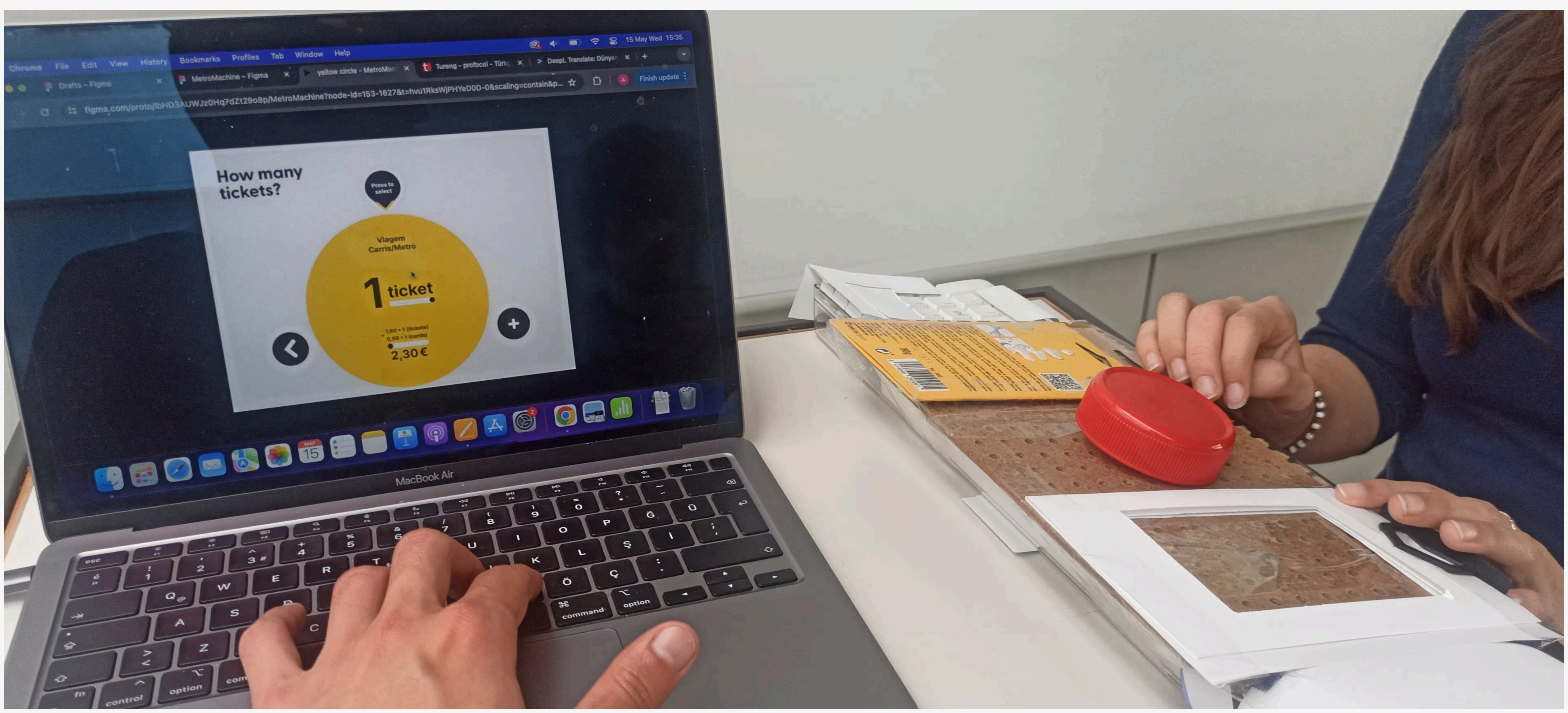
Watch one test session:

https://bit.ly/art-video-demo

Navegante machine

Public transportation should aim to be universal. However, for many the journey ends before it even begins: at the ticket machine. In the Lisbon metro, a screen is the input for this system, but for a blind user it is an impenetrable wall of glass. What happens to urban mobility when design forgets to be inclusive?

Navegante machine is a UX research project dedicated to redesign these machines in a more inclusive way. By moving away from purely visual interfaces, this study explored the **tactile and auditory dimensions of interaction**, developing two distinct solutions: a high-intuition rotary encoder and a familiar ATM-style audio-guided system. Through usability testing with blindfolded users and persona-driven flows, the project demonstrates that a simplified, multi-sensory information hierarchy doesn't just aid accessibility, it also **reduces friction and cognitive load for every passenger**, ensuring the city's transportation stays open for everyone.

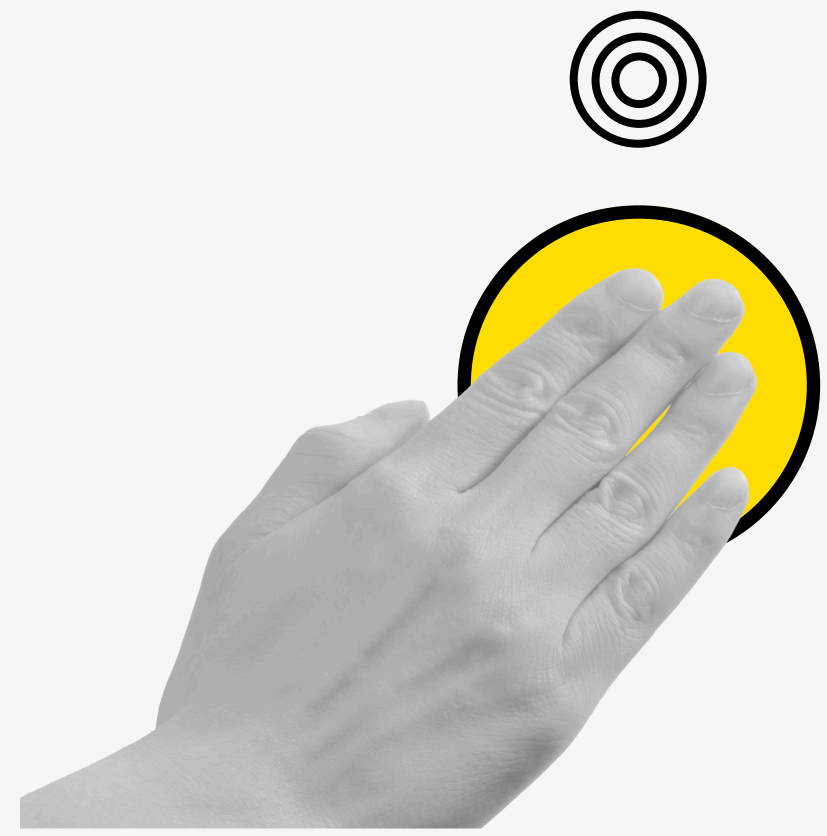


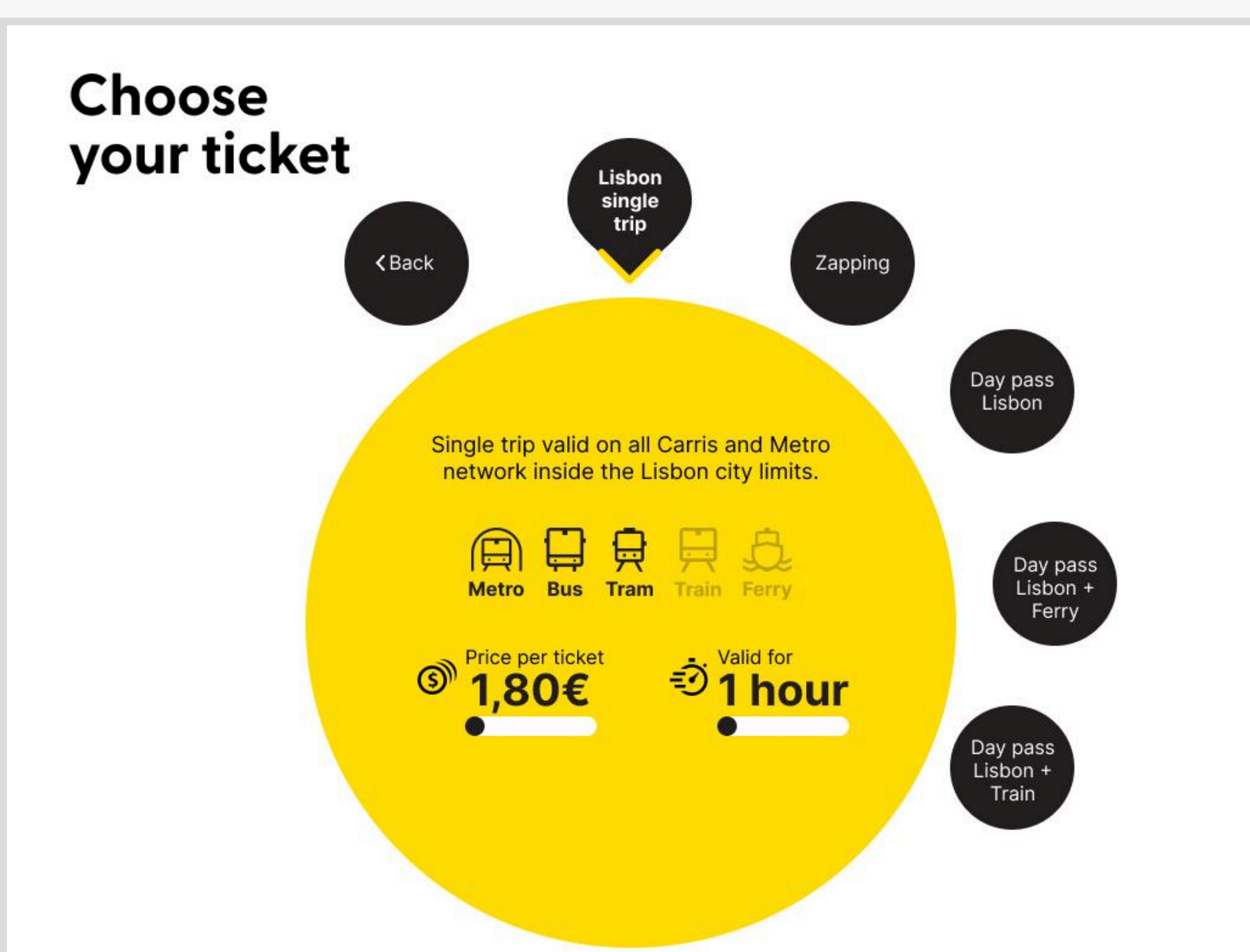
App's home screen on a mobile phone

Project information:

2024 | Group work | Interface prototype with a physical input for User testing

My roles: User research with interviews, UI design, Prototyping, User testing





The User interface is based on two main gestures: rotate to navigate throw the options, pressing to select. This makes it easy for blind users, as the physical input can easily be combined with audio feedback, and also makes it intuitive and simple for all the other passengers too.

rePraça

Public squares exist around the entire city; hundreds of citizens interact with them every day. However, what if those citizens were the planners of these places? Would these squares keep their current shape? How do one's dreams for this square differ from the one that currently exists, and how differently does each visitor imagine these places?

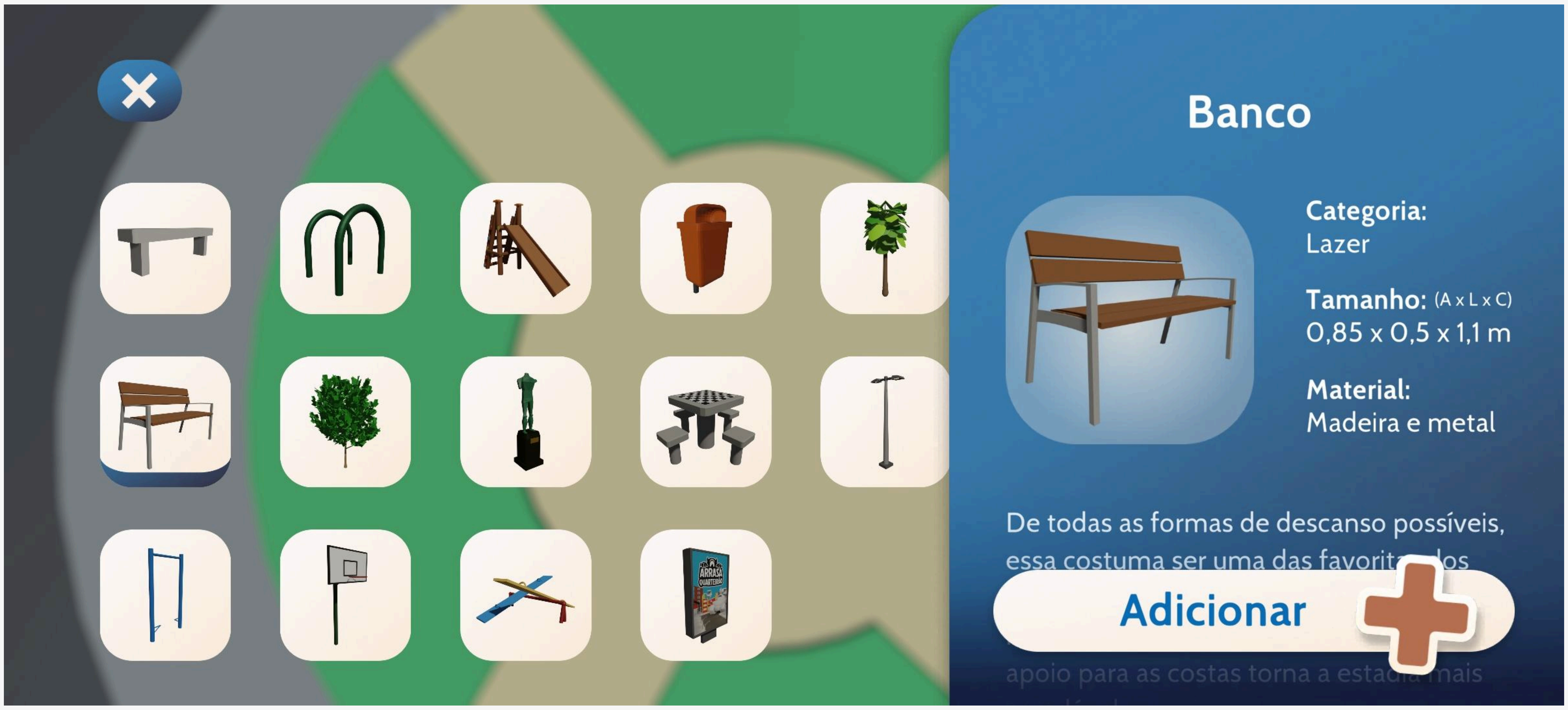
rePraça ("re-Square") is a virtual creator of city squares. This project aims to work the imagination of public spaces by its passersby; it stems from the desire to create a platform where citizens can create, share and exchange their views about city structures.

App's home screen on a mobile phone

Project information:

2022 | Individual work | Mobile app developed with Unity
Designed for the course Final Project - Digital Media Design

The perspective view of a virtual square, every mobilia item in the scene was added by the player. The game development used the Unity game engine.



Selection menu with every item available to be used in the design of your square; all the items were modeled with the Blender 3D software

View the project:

https://feliperpv.com/repraca/

GET IT ON Google Play

Arrasa Quarteirão

Arrasa Quarteirão ("Blockbuster") is an action-packed arcade game where you play as MAQ-11, a friendly blue monster on the run from a war tank controlled by scientists. Your goal is to demolish as many buildings as fast as you can!

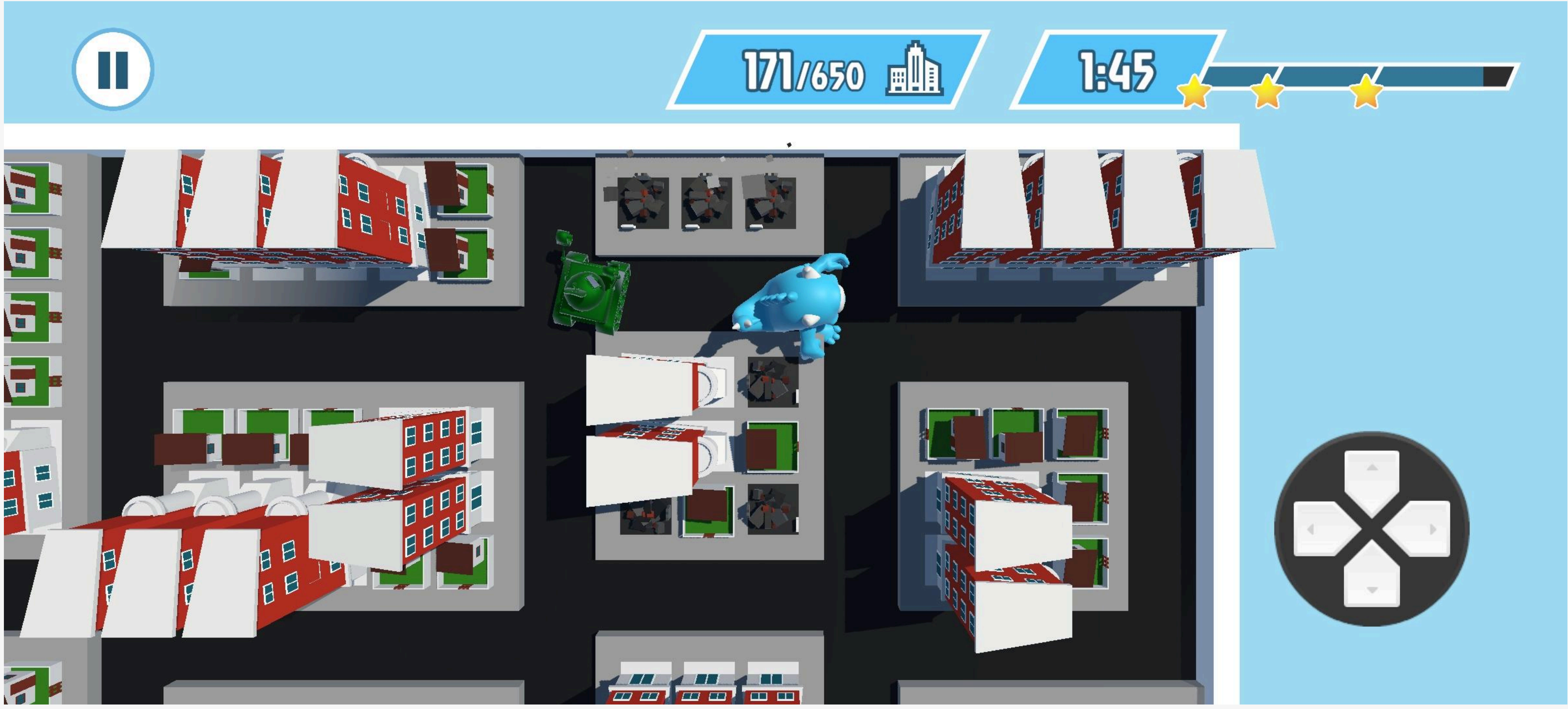
This project's development began in the Computing for Games course at PUC-Rio, extending later to many other classes, where new features were added and complexity grew. Since I worked in many different areas of the game, this project was a great lesson in teamwork and management of multiple teams.

Game home screen

Project information:

2021 - 2023 | Group work | Mobile game developed with Unity

My roles: Game design, user interface, coding, animation, sound design



Game screen during a level. The player controls the blue monster while running away from the green tank. Game HUD shows the score from destroying buildings and time left.

During gameplay, the pause screen features an illustration of the MAQ-11 character for a friendly and inviting appearance. This also reinforces the presence of the monster, as it is only seen from a top-down perspective while playing.

View the project:

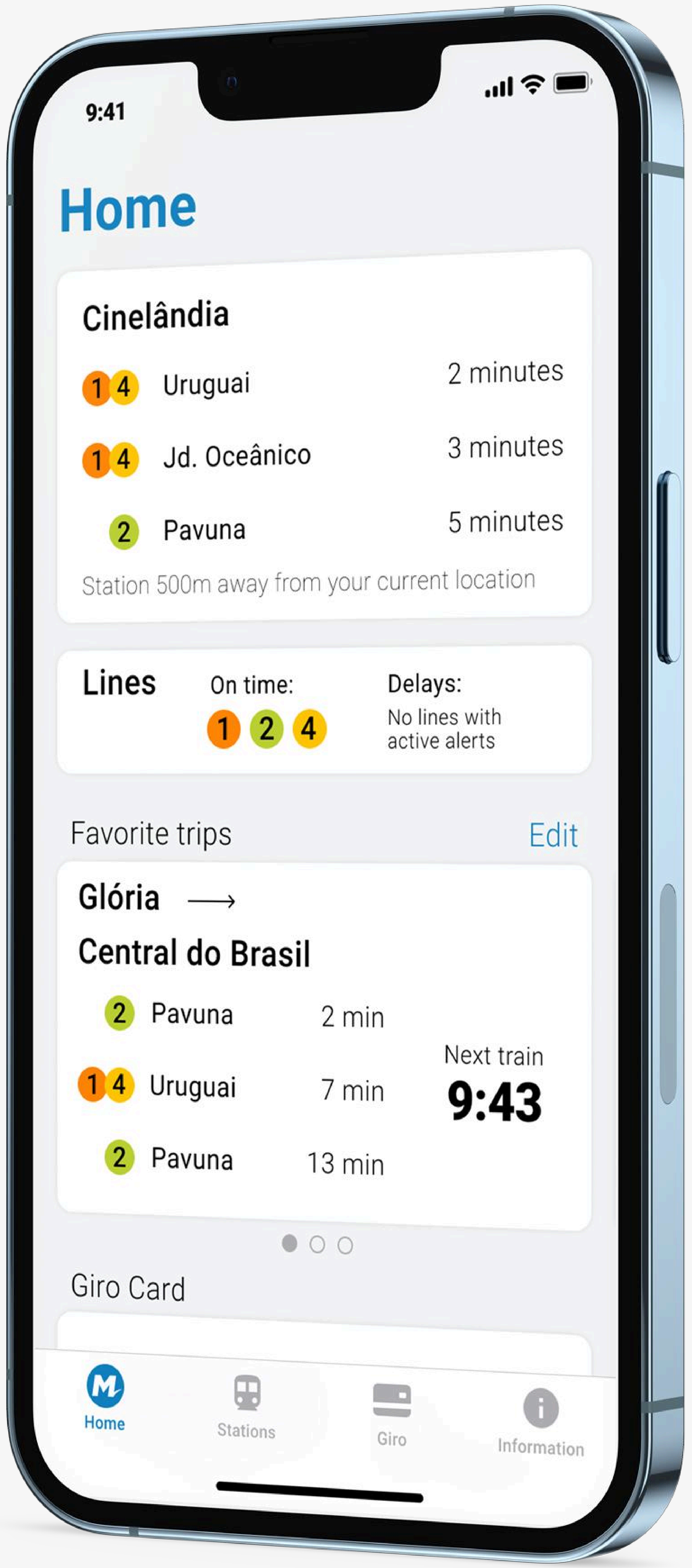
https://feliperpv.com/arrasaquarteirao/

GET IT ON Google Play

MetrôRio

Redesign project for the official Rio de Janeiro metro app. The idea originated after noticing the large number of taps required to check basic information about the transit itinerary.

In this prototype, the main focus was to minimize the number of interactions that happened with the interface to obtain the desired information. To achieve this, I created a new home screen that uses the user's geolocation to gather information about the nearest station and their favorite routes. In this way, it is possible to view the departure time of the next train without any additional steps.

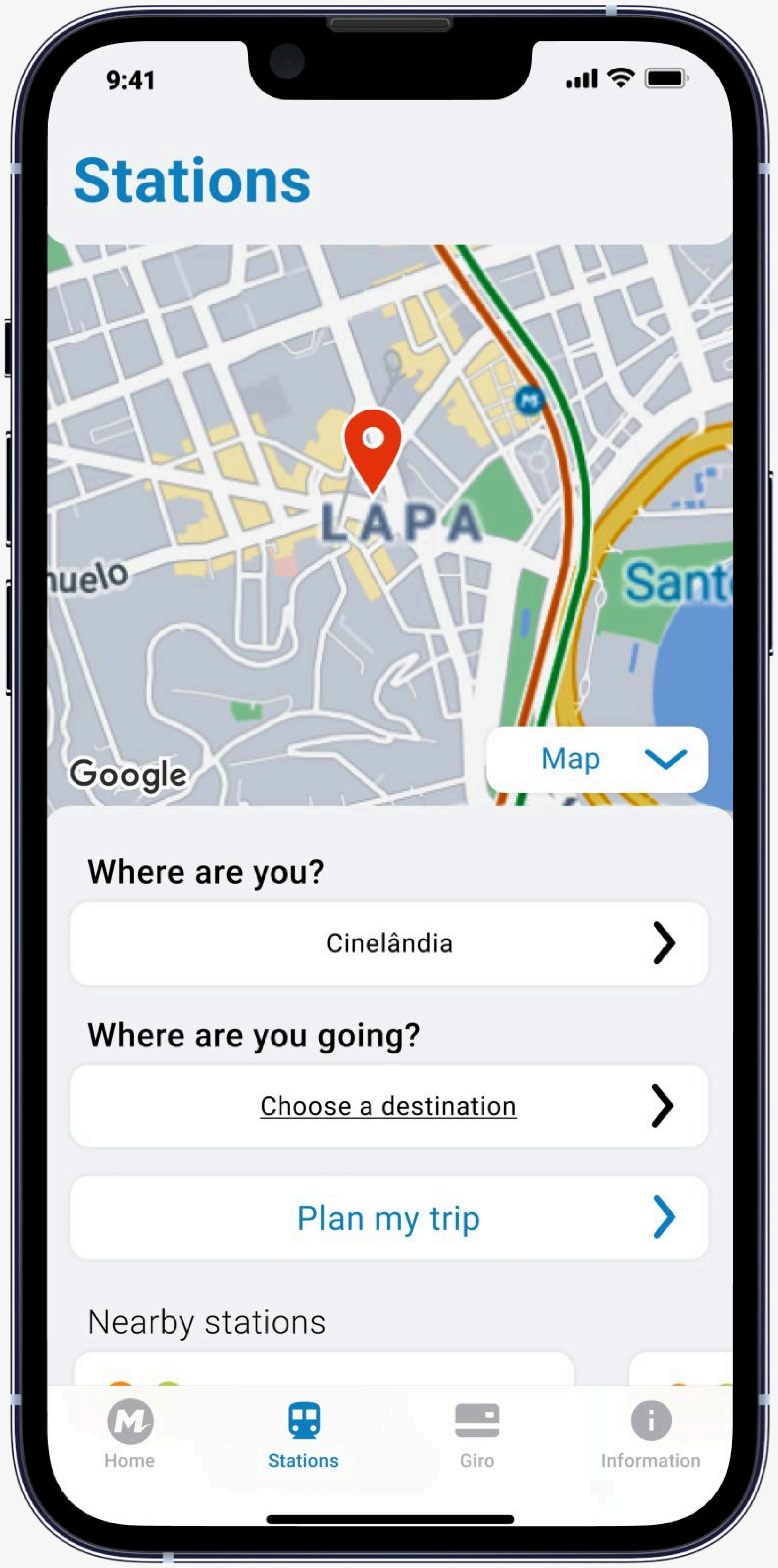


App home screen. The first widget displays the user's closest station information.

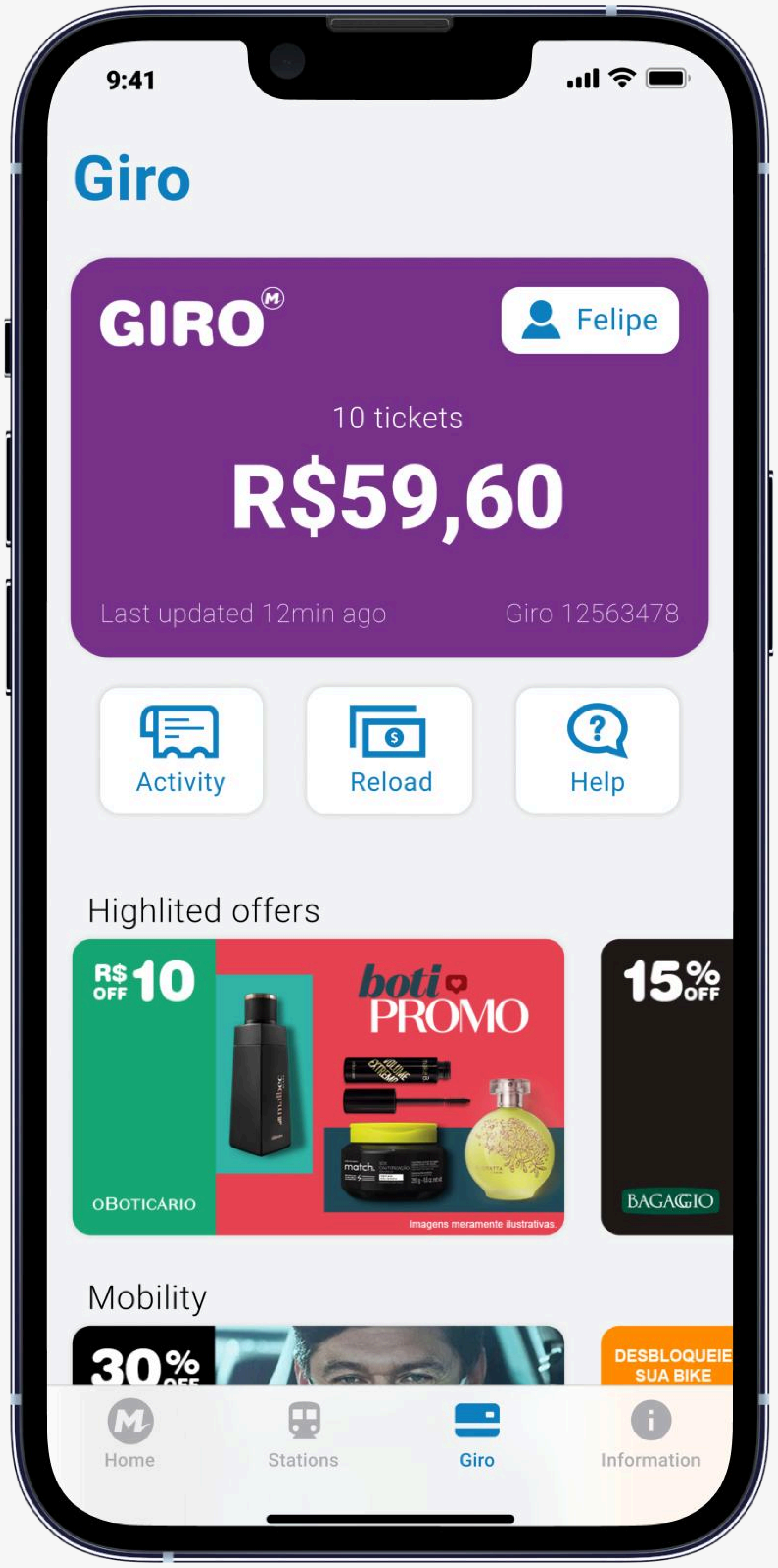
Project information:

2021 | Individual work | Figma prototype

Designed for the Virtual Prototyping course



Screen featuring all the metro stations. There is an interactive map and a route planner, where the "Where are you?" field is loaded automatically with the user's location.



Giro Card is the official payment system for the metro. This screen allows the user to check their balance and its corresponding amount of tickets, as well as to top up credit. Promotions and offers from partners are also displayed.

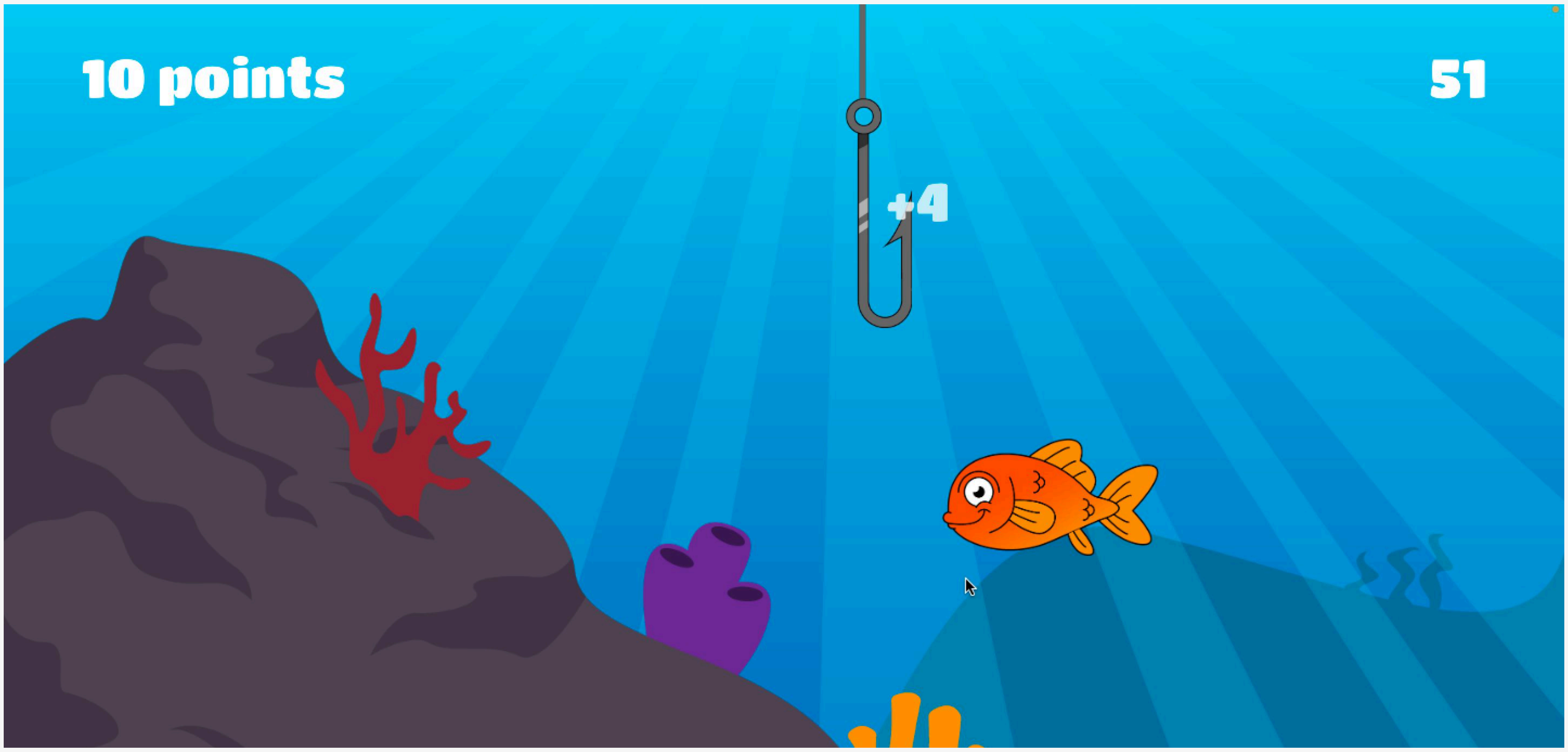
Open the prototype:

https://www.figma.com/proto/xMmnTxu0wys31CyodG4DbB/DSG1720-MetrôRio

Virtual Fishing

For our Physical and Logical Interfaces class, we had to develop an interactive experience using any of the various types of sensors available. We decided to do something fun, something that would only be possible with the Arduino logic board: using water to control a virtual game.

That is how "Virtual Fishing" was born, a game where you aim to catch as many fish as possible while avoiding obstacles that decrease your points. In just 60 seconds, see how many fish you can catch while dodging the garbage. Challenge your friends to see who can score the most points!

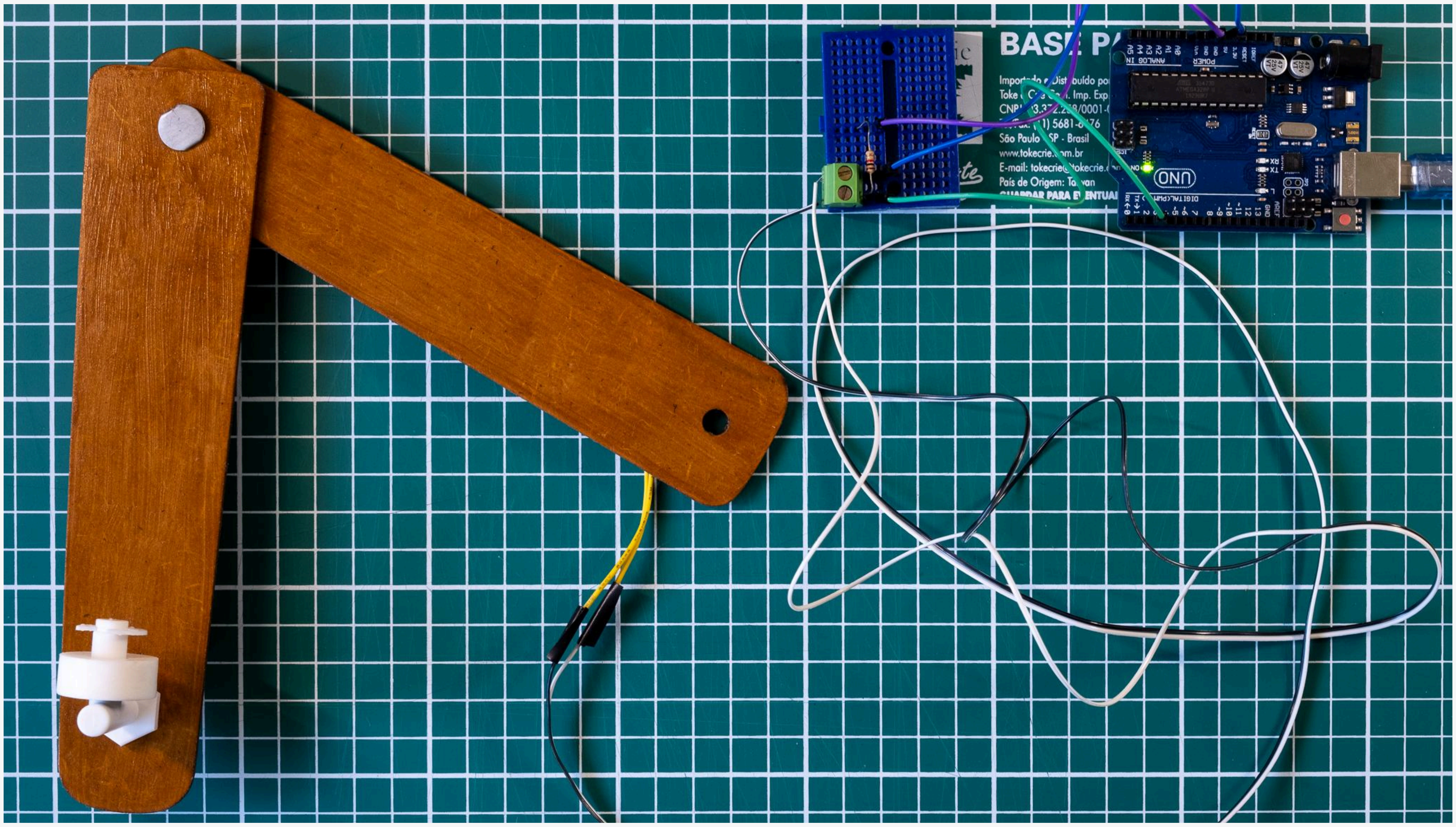


Main game screen. The player controls a hook to catch fish. The top of the screen displays the player's score and the time remaining in the game.

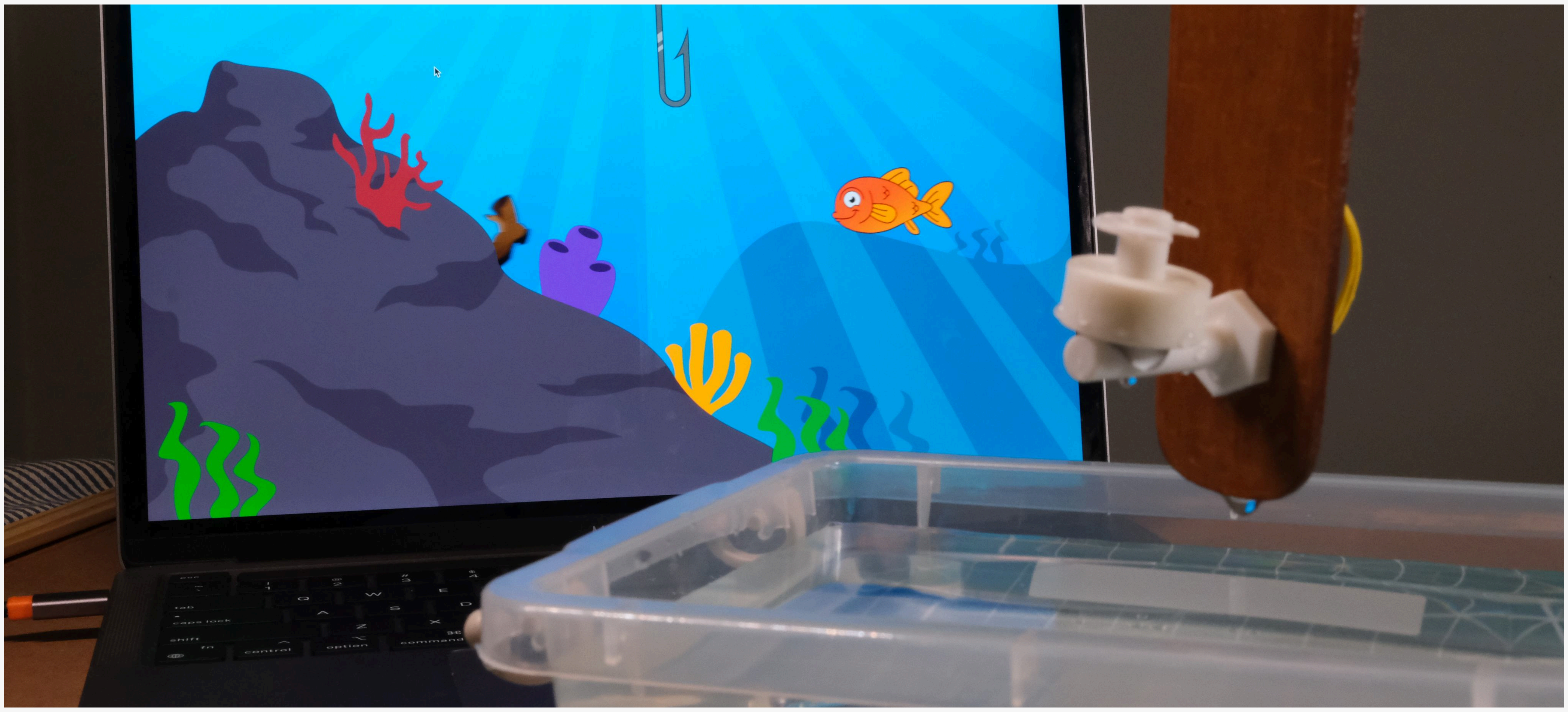
Project information:

2022 | Group work | Game developed with Processing and Arduino

My roles: Coding, Game design, hardware production



The water level sensor is attached to a wooden strip, forming the fishing rod used by the player. The Arduino connects the sensor to the computer running the game.



A fishing rod and a container of water are in front of the computer. When the sensor is out of the water, the hook on the game screen goes up.

Watch how it works:

https://vimeo.com/801798969

Learn more

felipeprpv.com